

A Summary of Two NLP Research Papers: Sentiment Analysis of Product Reviews and Fake News Detection

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Abstract—This document presents a combined summary of two research papers in the field of Natural Language Processing (NLP). The first paper proposes a sentiment analysis system that classifies e-commerce product reviews as positive, neutral or negative by fine-tuning the BERT transformer model and comparing it with TF-IDF and Bi-LSTM baselines. The second paper proposes a fake news detection system that classifies news articles as real or fake using classical machine learning, a hybrid CNN-LSTM network and a fine-tuned BERT model on the ISOT and LIAR datasets. For each paper, the problem statement, literature survey, methodology and key contributions are summarised. The two works are then compared, the NLP techniques common to both are explained, and the main challenges and future directions are discussed. The summary shows that both problems can be solved as text classification tasks and that pre-trained transformer models are the most promising approach for both.

Keywords—Natural Language Processing, Text Classification, Sentiment Analysis, Fake News Detection, BERT, LSTM, TF-IDF, Deep Learning

I. INTRODUCTION

Natural Language Processing (NLP) is the branch of Artificial Intelligence that enables computers to read, understand and generate human language. Every day, huge amounts of text are produced on e-commerce websites, news portals and social media, and NLP techniques are needed to turn this unstructured text into useful information. Text classification is one of the most important NLP tasks, where a piece of text is assigned to one of several predefined categories.

This document summarises two research papers that both apply text classification to real-world problems. The first paper, “Sentiment Analysis of Customer Product Reviews Using Transformer-Based Language Models”, classifies online product reviews as positive, neutral or negative. The second paper, “Fake News Detection Using Natural Language Processing and Deep Learning Techniques”, classifies news articles as real or fake. Both papers compare classical machine learning, recurrent and convolutional deep learning, and transformer-based models such as BERT [1].

Section II summarises the first paper and Section III summarises the second paper. Section IV compares the two works, Section V explains the NLP techniques common to both, Section VI discusses challenges and future scope, and Section VII concludes the summary.

II. PAPER 1: SENTIMENT ANALYSIS OF CUSTOMER PRODUCT REVIEWS

A. Problem Statement and Objectives

The rapid growth of e-commerce has changed the way people buy products. Before purchasing, most customers read the reviews written by earlier buyers, and sellers depend on the same reviews to understand what customers like or dislike about a product. Popular products can collect thousands of reviews, which makes manual analysis slow, costly and inconsistent. Sentiment analysis, also known as opinion mining, is the branch of Natural Language Processing (NLP) that automatically extracts the polarity of an opinion from text [2].

Early work treated sentiment analysis as a standard text classification problem and applied Naive Bayes, Maximum Entropy and Support Vector Machine classifiers on word-count features [3]. These methods are fast and simple but ignore word order, so sentences such as “not bad at all” or “I expected much better” are often misclassified. Deep learning models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks improved results by learning features from word embeddings [4]–[7]. More recently, transformer-based language models pre-trained on large text corpora, such as BERT [1], have become the standard approach for most NLP tasks because they model the full context of every word in both directions.

The main objectives of this work are:

- To build a clean, reproducible preprocessing pipeline for noisy product review text.
- To fine-tune a pre-trained BERT model for three-class (positive, neutral, negative) review sentiment classification.
- To compare the transformer model with classical machine learning and Bi-LSTM baselines using standard evaluation metrics.
- To design a simple interface through which a user can enter a review and obtain its predicted sentiment with a confidence score.

B. Literature Survey

Pang et al. [3] were among the first to apply machine learning to sentiment classification of movie reviews and showed that unigram features with SVM performed better than hand-crafted rules. Maas et al. [8] released the widely used IMDb Large Movie Review dataset and learned word vectors that capture sentiment information. Word embedding techniques such as Word2Vec [4] and GloVe [5] represent words as dense vectors so that semantically similar words lie close to each other, which became the input layer for most neural models.

Kim [6] showed that a simple CNN with one convolution layer over pre-trained word vectors achieves strong results

on several sentence classification benchmarks. LSTM networks [7] handle long-range dependencies in text and have been widely used for review sentiment analysis. Socher et al. [9] introduced the Stanford Sentiment Treebank and demonstrated that modelling sentence structure helps to capture negation.

The transformer architecture proposed by Vaswani et al. [10] replaced recurrence with self-attention and allowed efficient training on very large corpora. Devlin et al. [1] introduced BERT, which is pre-trained using masked language modelling and can be fine-tuned for downstream tasks by adding a single output layer. RoBERTa [11] improved BERT through better pre-training choices, while DistilBERT [12] reduced model size for faster inference. McAuley and Leskovec [13] collected a large Amazon product review corpus that is commonly used for review-level research. Based on this survey, fine-tuned transformer models are selected as the core of the proposed system, with classical and recurrent models as baselines.

C. Proposed Methodology

1) *Dataset*: The system uses publicly available Amazon product reviews [13]. Each review contains the review text and a star rating from 1 to 5. Ratings of 1–2 are labelled negative, 3 is labelled neutral and 4–5 are labelled positive. A balanced subset is sampled so that each class is equally represented, and the data is split into training (80%), validation (10%) and test (10%) sets.

2) *Text Preprocessing*: Raw review text contains HTML tags, URLs, emojis, repeated characters and spelling mistakes. The preprocessing steps are: (i) removal of HTML tags and URLs, (ii) lower-casing (for baseline models), (iii) expansion of contractions such as “don’t” to “do not” so that negation is preserved, (iv) tokenisation, and (v) stop-word removal and lemmatisation for the classical baselines only. For BERT, the WordPiece tokenizer is applied directly and stop words are kept, because the model relies on the full sentence context.

3) *Baseline Models*: Two classical baselines are trained on TF-IDF features of unigrams and bigrams: Multinomial Naive Bayes and Logistic Regression. A deep learning baseline uses a two-layer Bidirectional LSTM with 300-dimensional GloVe [5] embeddings followed by a dense softmax layer.

4) *Transformer Model*: The main model is the pre-trained bert-base-uncased model [1], which has 12 encoder layers, 768 hidden units and 12 attention heads. Each review is converted to the form [CLS] tokens [SEP], truncated or padded to 256 tokens. The final hidden state of the [CLS] token is passed through a dropout layer and a linear layer with softmax activation to produce probabilities for the three classes. The complete network is fine-tuned end-to-end using the AdamW optimiser with a small learning rate, linear warm-up and cross-entropy loss. Early stopping on the validation F1-score is used to prevent over-fitting.

5) *Evaluation Metrics*: All models are evaluated on the same held-out test set using accuracy, precision, recall and macro-averaged F1-score. A confusion matrix is used to study the errors between neighbouring classes, especially between neutral and the two polar classes, which is known to be the most difficult case.

D. Key Contributions

- A complete pipeline for noisy e-commerce review text, including negation-aware preprocessing.
- Fine-tuning of BERT for three-class review sentiment classification.
- A fair comparison of TF-IDF, Bi-LSTM and transformer models on the same data split.

III. PAPER 2: FAKE NEWS DETECTION USING NLP AND DEEP LEARNING

A. Problem Statement and Objectives

News consumption has moved from newspapers and television to social media platforms and messaging applications. While this makes information easily accessible, it also makes it easy for anyone to publish content without editorial checks. Fake news is defined as news articles that are intentionally and verifiably false and could mislead readers [14]. A large-scale study of Twitter by Vosoughi et al. [15] found that false news spreads farther, faster and deeper than true news, which shows how serious the problem has become.

Professional fact-checking organisations verify claims manually, but this process is slow and does not scale to the huge amount of content published every day. Therefore, automatic detection methods are required. Since the content of a news article is written in natural language, NLP techniques can be used to learn the linguistic patterns, writing style and word usage that differ between fake and genuine news [16]. Fake articles often use emotional words, exaggerated headlines, fewer sources and a sensational tone, and these signals can be captured by machine learning and deep learning models.

The main objectives of this work are:

- To study existing approaches and benchmark datasets for fake news detection.
- To design an NLP preprocessing and feature extraction pipeline for news articles.
- To implement and compare classical machine learning, CNN-LSTM and BERT-based classifiers for real/fake news classification.
- To provide a simple web interface in which a user can paste a news article and receive a prediction with a confidence score.

B. Literature Survey

Shu et al. [14] presented a detailed survey of fake news detection from a data mining perspective and divided the approaches into news-content-based and social-context-based methods. Wang [17] released the LIAR dataset containing about 12.8K short political statements labelled with six degrees of truthfulness, and showed that a hybrid CNN using text and metadata performs better than text-only models. Pérez-Rosas et al. [16] built datasets covering several news domains and studied linguistic features such as n-grams, punctuation, readability and psycholinguistic categories for fake news detection.

Ahmed et al. [18] used n-gram features with TF-IDF weighting and compared six classical classifiers, finding that linear models such as Linear SVM work well; their ISOT dataset is widely used in student and research projects. Shu et al. [19] introduced FakeNewsNet, which includes news content along with social context from PolitiFact and GossipCop. Ruchansky et al. [20] proposed CSI, a hybrid

deep model that combines the text of articles, the response of users and the behaviour of sources.

Deep learning models remove the need for manual feature engineering. CNNs [6] capture local n-gram patterns and LSTMs [7] capture long-range dependencies in text. Transformer-based models such as BERT [1] learn deep contextual representations and have achieved state-of-the-art results on many NLP tasks. Kaliyar et al. [21] proposed FakeBERT, which combines BERT embeddings with parallel one-dimensional convolution blocks for fake news classification. Based on this survey, the proposed work compares content-based classical, hybrid deep learning and transformer models on the same datasets.

C. Proposed Methodology

1) *Datasets*: Two public datasets are used: the ISOT Fake News dataset [18], which contains full articles labelled as real or fake, and the LIAR dataset [17], which contains short statements. For LIAR, the six labels are mapped into two classes (true, mostly-true and half-true as real; barely-true, false and pants-fire as fake). Each dataset is split into 80% training, 10% validation and 10% test data using stratified sampling.

2) *Preprocessing and Feature Extraction*: The headline and body are concatenated to form the input text. The preprocessing steps include removal of URLs, HTML tags, special characters and extra spaces, lower-casing, tokenisation, stop-word removal and lemmatisation. Source names and agency tags such as “(Reuters)” are removed so that the model cannot learn a shortcut from the publisher name. For classical models, TF-IDF [22] vectors of unigrams and bigrams are extracted. For deep learning models, the text is converted to integer sequences and padded to a fixed length.

3) *Classical Machine Learning Models*: Logistic Regression, Multinomial Naive Bayes, Linear Support Vector Machine and Random Forest classifiers are trained on TF-IDF features. Hyper-parameters are selected using grid search with 5-fold cross-validation on the training set.

4) *Hybrid CNN-LSTM Model*: The input sequence is passed through an embedding layer initialised with pre-trained GloVe vectors. A one-dimensional convolution layer with max-pooling extracts local phrase features, and an LSTM layer then models the order of these features across the article. The output is passed through a dropout layer and a dense layer with sigmoid activation to predict the probability of the article being fake.

5) *BERT-Based Model*: The pre-trained bert-base-uncased model [1] is fine-tuned for binary classification. The article is tokenised with the WordPiece tokenizer and truncated to 512 tokens. The [CLS] representation is fed to a classification head, and the whole model is trained using the AdamW optimiser and binary cross-entropy loss.

6) *Evaluation*: All models are compared on the same test sets using accuracy, precision, recall, F1-score and the ROC-AUC. Since wrongly marking fake news as real is more harmful, recall of the fake class is given special importance.

D. Key Contributions

- A content-based fake news classifier evaluated on both long articles (ISOT) and short statements (LIAR).

- Removal of publisher cues so that the model learns from language, not from the source name.
- Comparison of classical, hybrid CNN-LSTM and BERT models with special focus on recall of the fake class.

IV. COMPARATIVE ANALYSIS

TABLE I
COMPARISON OF THE TWO RESEARCH PAPERS

Aspect	Paper 1: Sentiment Analysis	Paper 2: Fake News Detection
Task	Multi-class (positive / neutral / negative)	Binary (real / fake)
Input text	Short product reviews	News headline and article body
Datasets	Amazon product reviews	ISOT and LIAR
Classical models	Naive Bayes, Logistic Regression	LR, Naive Bayes, Linear SVM, Random Forest
Deep learning model	Bi-LSTM with GloVe	Hybrid CNN-LSTM with GloVe
Transformer	BERT-base, 256 tokens, 3-class head	BERT-base, 512 tokens, binary head
Main metric	Macro F1-score	F1-score and recall of fake class
Main challenge	Neutral class, negation, sarcasm	Topic/source bias, evolving writing style
Application	Summarising customer feedback	Supporting fact-checkers

Table I shows that the two papers follow the same overall research design: a clean preprocessing pipeline, simple TF-IDF baselines, a neural baseline using pre-trained word embeddings and finally a fine-tuned transformer. The main difference lies in the nature of the text. Reviews are short and opinion-heavy, so capturing negation and intensity is important. News articles are long and factual in style, so the model must capture writing style and deception cues across many sentences, which is why Paper 2 uses a longer input length and a CNN-LSTM hybrid.

Another difference is the cost of errors. In sentiment analysis, confusing neutral and positive reviews has limited impact, so macro F1-score is a suitable summary metric. In fake news detection, a fake article that is predicted as real can mislead many readers [15], so recall of the fake class is treated as a priority.

V. COMMON NLP TECHNIQUES USED

A. Text Preprocessing

Both works clean the raw text by removing HTML tags, URLs and special characters, then tokenise it. Stop-word removal and lemmatisation are applied only for classical models, because transformer models need the complete sentence to understand context.

B. TF-IDF Features

Term Frequency-Inverse Document Frequency [22] gives a high weight to words that are frequent in a document but rare in the whole collection. Unigram and bigram TF-IDF vectors with linear classifiers form fast and strong baselines [3], [18].

C. Word Embeddings

Word2Vec [4] and GloVe [5] map every word to a dense vector so that words with similar meaning have similar vectors. These vectors are used as the first layer of the LSTM and CNN models in both papers.

D. CNN and LSTM Networks

A CNN [6] slides filters over the word sequence to detect important phrases, while an LSTM [7] reads the sequence step by step and remembers long-range information using gates. Paper 1 uses a Bi-LSTM, and Paper 2 combines both into a CNN-LSTM.

E. Transformers and BERT

The transformer [10] uses self-attention so that every word can directly attend to every other word. BERT [1] is pre-trained on large corpora with masked language modelling and is then fine-tuned with a small classification head. Both papers use the [CLS] token output of bert-base-uncased for classification.

VI. CHALLENGES AND FUTURE SCOPE

The following challenges are common to both works:

- Sarcasm, irony and code-mixed (Hindi-English) text are difficult for current models.
- Transformer models need GPUs and are slow on large volumes; smaller models such as DistilBERT [12] can reduce this cost.
- Models may learn shortcuts from dataset bias, such as publisher names or product categories, instead of real language patterns.
- Fake news styles change over time, so the detection model must be retrained regularly.

Future work includes aspect-based sentiment analysis, which finds opinions about specific product features such as battery or camera; adding social-context and user-behaviour features for fake news detection [19], [20]; multilingual models for Indian languages; and explainability methods that highlight the words responsible for each prediction.

VII. CONCLUSION

This document summarised two NLP research papers on sentiment analysis of product reviews and on fake news detection. Although the applications are different, both papers treat the problem as text classification and follow a common path from classical TF-IDF models to deep learning and finally to fine-tuned transformer models such as BERT. The study shows that pre-trained language models are the most promising approach for understanding context, negation and writing style, while classical models remain useful as fast and interpretable baselines. Together, these papers show how NLP can help both businesses and society make better use of online text.

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